Earth-Based Repeat-Pass SAR Interferometry of the Moon: Spatial–Temporal Baseline Analysis

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***Abstract*— Earth-based repeat-pass synthetic aperture radar (SAR) interferometry (InSAR) is a powerful tool to obtain the high-precision lunar terrain and deformation information, due to the short revisit period as well as the all-weather, long-time, and large-coverage characteristics of radar lunar observation. In this article, the baseline formation mechanism is investigated. Theoretical analyses show that the interferometric baselines mainly come from the lunar libration; however, the lunar libration may also result in a significant change of baselines and further lead to serious geometric decorrelation. To address this problem and ensure the Earth-based repeat-pass InSAR of the Moon, the precise relative motion model between the radar and the lunar target, considering lunar libration, is established, and the subradar point (SRP) wagging phenomenon is analyzed. Then, to evaluate the effects of SRP wagging on InSAR coherence, a geometric decorrelation model based on azimuth angle and incident angle is proposed. Based on that, the spatial–temporal baseline for Earth-based repeat-pass InSAR of the Moon is analyzed based on numerical simulation. Results show that the temporal baseline has obvious periodicity with a cycle of about 27 days. When the temporal baseline is an integral multiple of 27, the spatial baseline usually has the minimum value and the geometric decorrelation effect is relatively slight. Moreover, using the geometric decorrelation criteria, we examine the effects of radar frequency, position, and target location on the quantity of viable interferometric pairs, which has guiding significance for the selection of radar system parameters, radar position, and lunar target area.**

***Index Terms*— Earth-based radar, interferometry, Moon, repeat-pass, spatial and temporal baseline.**

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I. Introduction

THE Moon is Earth's only natural satellite, the nearest celestial body to our Earth and the only extraterrestrial planet that human beings can reach at present [1]. In 1946, the Evans Signal Laboratory (ESL) successfully received the radar echo reflected from the Moon, which opened the chapter of human exploration of the Moon with Earth-based radar [2]. Since then, Earth-based radar has never stopped transmitting electromagnetic waves to the Moon and many technologies, such as delay-Doppler imaging [3]–[7] and radar interferometry [8]–[11], have been developed to obtain high-resolution images and topography of the lunar surface.

During the past decades, the Earth-based radar has been successfully proved as a powerful tool to observe the Moon on Earth. The first global radar mapping results for the lunar nearside were achieved in the 1980s, which was used to assist in landing-site selections for the Apollo missions [5]. Nowadays, the resolution of Earth-based radar imaging on the Moon can reach the meter level, which is equivalent to that of a lunar probe [11]. Moreover, radar interferometry techniques have also been applied to obtain the lunar surface topography, especially the poles, which is helpful to search for ice deposits in permanently shadowed craters on the lunar surface [12]–[14]. In 1972, Shapiro [8] and Zisk [9] obtained measurements of lunar topography for the Alphonsus–Arzachel region and the Crisium area. The spatial resolution was 1–2 km and the elevation accuracy was on the order of 500 m. The interferometer was composed of the Haystack–Westford radar systems and the interferometric baseline was only 1200 m [8], [9]. In 2000, Margot [10] got the highest resolution topographic maps of the lunar surface by the Earth-based radar interferometry technology at that time. The obtained digital elevation models (DEMs), achieved by goldstone solar system radar (GSSR), were at 150-m spatial resolution and 50-m vertical accuracy [10], which was proven pivotal in the Lunar Prospector impact planning. In 2006, with the improvement of observation geometry, Hensley *et al.* [15] generated a topographic map of the south polar region of the Moon with more than three times finer spatial resolution and ten times finer vertical accuracy than before. Then, in 2009, since GSSR has been improved to allow 40-MHz bandwidth imaging, Hensley achieved lunar topographic maps at 15-m spatial resolution with about the same height accuracy, which is the best result obtained by Earth-based radar so far [15], [17].

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Current Earth-based radar interferometric results are all obtained through single-pass mode. For example, in GSSR, the 70-m parabolic dish antenna was used to transmit the signal, and two 34-m antennas separated by 13 km formed the receiving interferometer [18]. The altitude of ambiguity [19], defined as the altitude difference that generates an interferometric phase change of 2π, is about 500 m in the GSSR configuration. The advantage of single-pass mode is that the data have good coherence and little atmospheric effect [20]. However, the interferometric baseline, decided by the positions of two receiving antennas, is usually very short, which limits the accuracy of elevation information transformed from the interferometric phase [21]. What is more, since the primary and secondary images are all obtained simultaneously in single-pass mode, it cannot measure deformation [22].

The Moon was previously thought to be fairly geologically quiet. Recently, Watters *et al.* [23] found moonquakes in the 1960s and 1970s occurred near fault scarps, and the proximity of moonquakes to the young thrust faults together with evidence of regolith disturbance and boulder movements on and near the fault scarps strongly suggests that the Moon is tectonically active. Impact craters are the most common surface features on the Moon. On March 17, 2013, a new crater, about 18.8 m in diameter, was recorded by the images from the Lunar Reconnaissance Orbiter Camera (LROC) and more than 200 related surficial changes (e.g., landslide) up to 30-km distant [24]. In addition, the lunar tidal deformation, a bulge about 51 cm high, caused by the Earth has been detected by NASA's Lunar Orbiter Laser Altimeter (LOLA) [25]. The measurement of this deformation puts forward high requirements for the instrument.

Fortunately, due to the well-known tidal locking effect [26], the revisit period of Earth-based radar for the near side of the Moon is only one day. This makes repeat-pass synthetic aperture radar (SAR) interferometry (InSAR) measurement possible. Thus, the Earth-based repeat-pass InSAR of the Moon is proposed here to form the observation baseline and obtain the topography and deformation information of the Moon. However, different from the spaceborne SAR maintained within a fixed orbital tube to ensure ground-track coverage repeatability [27], the relative track between the Earth-based radar and the lunar target cannot be manually controlled. Moreover, the relative motion between the Earth-based radar and the lunar target is very complex. It is necessary to consider not only the Earth rotation, the Moon rotation, and Moon revolution but also the high-order motions such as lunar libration and orbital precession [28]–[31]. The baseline formation mechanism of Earth-based repeat-pass InSAR of the Moon is not clear.

To address the aforementioned problems, in this article, the precise relative motion model between the radar and the lunar target, considering lunar libration, is established, and the subradar point (SRP) wagging phenomenon is analyzed. We find that the interferometric baselines mainly come from the lunar libration; however, the lunar libration may also result in a significant change of baselines and further lead to serious geometric decorrelation, which brings severe challenges to repeat-pass interferometry. Then, to evaluate the effects of SRP wagging on InSAR coherence, a geometric decorrelation model based on azimuth angle and incident angle is proposed. Based on that, the spatial–temporal baseline for Earth-based repeat-pass InSAR of the Moon is analyzed based on numerical simulation. Moreover, using the geometric decorrelation criteria, we examine the effects of radar frequency, position, and target location on the quantity of viable interferometric pairs, which has guiding significance for the selection of radar system parameters, radar position, and lunar target area.

TABLE I
SIX ORBITAL ELEMENTS OF THE MOON

Orbital element	Value	Symbol
Eccentricity	0.0549	e
Semimajor axis	384400 km	a
Inclination	18.28°~28.58°	i
Right ascension of the ascending node (RAAN)	0°~360°	Ω
Argument of perigee	0°~360°	ω
True anomaly	0°~360°	v

This article is organized as follows. Section II describes the Earth-based radar observation geometry model and the SRP motion trajectory. In Section III, the geometric decorrelation model based on azimuth and incident angle is constructed, the coherence is analyzed in detail, and the corresponding spatial–temporal is also calculated. Section IV provides the simulation result and the statistical characteristic of the spatial–temporal baseline. The effects of different key parameters on Earth-based repeat-pass InSAR are also summarized. Section V concludes this article and provides a vision for Earth-based repeat-pass of the Moon.

II. EARTH-BASED RADAR OBSERVATION GEOMETRY MODEL

The geometric model is the basis of Earth-based radar imaging and interferometric analysis of the Moon, and the key is to establish the relative motion model between the radar and target. In this section, the motion relationship between the Earth and the Moon is introduced and the motion of the SRP is described considering three kinds of motion, namely, Earth rotation, Moon rotation, and Moon revolution.

A. Relative Motion Between the Earth and the Moon

The Earth–Moon geometry is shown in Fig. 1. Actually, the relative motion between the Earth and the Moon can be modeled as a classical two-body problem, and six orbital elements can be correspondingly adopted to describe the Moon's trajectory, as shown in Table I. It should be noted that both the eccentricity and semimajor given in Table I are mean values, and they are subject to periodic variations due to solar perturbations. Moreover, every 18.6 years, the inclination angle reaches a maximum value of 28.58°, which is the sum of Earth's equatorial tilt (23.43°) and the Moon's orbital inclination (5.15°) to the ecliptic; then, 9.3 years later, the inclination angle reaches its minimum of 18.28°. With respect

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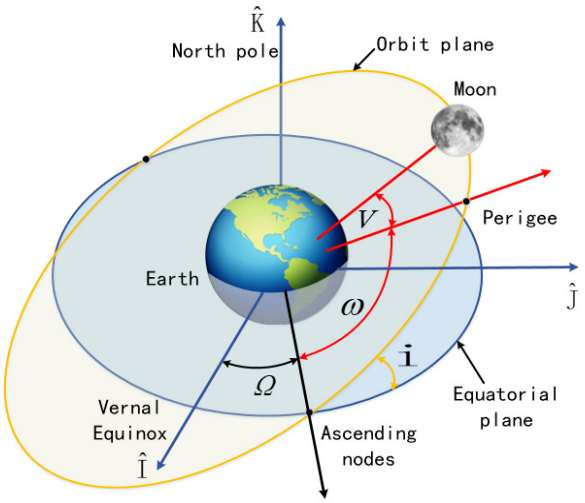


Fig. 1. Orbit of the Moon.

to an Earth equator and equinox referenced coordinate system, right ascension of the ascending node (RAAN), argument of perigee, and true anomaly have change periods of 18.6 years, 8.85 years, and 27.32 days, respectively [26].

Then, the Cartesian coordinate of the Moon in the Earth-centered inertial (ECI) coordinate frame $(X_{\text{moon}}, Y_{\text{moon}}, Z_{\text{moon}})$ can be calculated from the six orbital elements

$$\begin{bmatrix} X_{\text{Moon}} \\ Y_{\text{Moon}} \\ Z_{\text{Moon}} \end{bmatrix}_{\text{ECI}} = R_3(-\boldsymbol{\Omega})R_1(-\mathbf{i})R_3(-\boldsymbol{\omega}) \begin{bmatrix} a_p \cos \boldsymbol{v} \\ a_p \sin \boldsymbol{v} \\ 0 \end{bmatrix} \tag{1}$$

where

$$a_p = \frac{\boldsymbol{a}\left(1-\boldsymbol{e}^2\right)}{1+\boldsymbol{e}\cos \boldsymbol{v}} \tag{2}$$

and the rotation matrices R_1, R_2, and R_3 are right-handed rotations of frame orientations defined by

$$R_1(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & \sin\alpha \\ 0 & -\sin\alpha & \cos\alpha \end{bmatrix} \tag{3}$$

$$R_2(\alpha) = \begin{bmatrix} \cos\alpha & 0 & -\sin\alpha \\ 0 & 1 & 0 \\ \sin\alpha & 0 & \cos\alpha \end{bmatrix} \tag{4}$$

$$R_3(\alpha) = \begin{bmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{5}$$

Actually, (1)–(5) can only describe the precise spatial position of the Moon center relative to the Earth center; however, the radar is not situated on the center of the Earth and the target is also not on the center of the Moon, and there is an additional apparent displacement of the coordinates, which should be considered in the construction of relative motion model between the radar and the lunar surface target. Moreover, as shown in Fig. 2, the Earth-based radar and lunar surface target are usually described by different coordinate frames. Thus, to establish the motion model of the radar relative to the target accurately and simplify the following

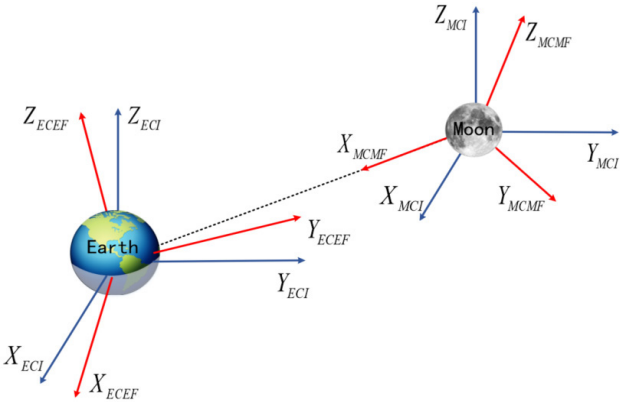


Fig. 2. Coordinates system in the Earth–Moon system.

analysis, it is necessary to convert the coordinates to the same coordinate frame, and the Moon-centered Moon-fixed (MCMF) coordinate system is adopted here. It should be noted that the conversions from other coordinate systems to the MCMF system, which are shown in Fig. 3, require the following six coordinate systems [30]–[33].

1) *Geodetic Coordinate System:* In this coordinate system, Earth's surface is approximated by an ellipsoid, and the radar locations near the surface are described in terms of latitude (φ_r), longitude (λ_r), and height (h_r).
2) *Earth-Centered Earth-Fixed Coordinate System:* The origin is the Earth's mass center. The x-axis passes through the Equator and Prime Meridian intersection. The z-axis passes through the north pole, and the y-axis is orthogonal to the xz plane.
3) *Earth-Centered Inertial Coordinate System:* The origin is the Earth's mass center. The xy plane coincides with the Earth's equatorial plane. The z-axis lies at a 90° angle to the equatorial plane and passes through the north pole.
4) *Moon-Centered Inertial Coordinate System:* The origin is the lunar mass center, and the x-, y-, and z-axes are parallel to the ECI coordinate system.
5) *Moon-Centered Moon-Fixed Coordinate System:* The origin is located at the lunar mass center, the x-axis points to the intersection of the lunar primary meridian and the lunar equator, the z-axis points to the lunar north pole, and the y-axis is orthogonal to the xz plane.
6) *Selenographic Coordinate System:* It is similar to the geodetic coordinate system (GCS) coordinate system. The Moon is regarded as a perfect sphere and the position of the lunar target is described as latitude (φ_t), longitude (λ_t), and height (h_t).

Based on these six coordinate systems, all the motions will be transformed to the MCMF coordinate system with the application of ephemeris data and detailed coordinate transformations. First, the coordinate of the Earth-based radar in Earth-centered Earth-fixed (ECEF) (X_r, Y_r, Z_r) can be obtained by

$$\begin{bmatrix} X_r \\ Y_r \\ Z_r \end{bmatrix} = \begin{bmatrix} (N(\varphi_r)+h)\cos\varphi_r\cos\lambda_r \\ (N(\varphi_r)+h)\cos\varphi_r\sin\lambda_r \\ \left(\frac{b^2}{a^2}N(\varphi_r)+h\right)\sin\varphi_r \end{bmatrix} \tag{6}$$

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where

$$N(\varphi_r) = \frac{m^2}{\sqrt{m^2 \cos^2 \varphi_r + b^2 \sin^2 \varphi_r}} \tag{7}$$

and m and b are the equatorial radius and the polar radius, respectively. Then, the coordinate of the Earth-based radar in MCMF (X_m, Y_m, Z_m) can be obtained by

$$\begin{bmatrix} X_m \\ Y_m \\ Z_m \end{bmatrix} = R_{\text{MCI}}^{\text{MCMF}} \cdot R_{\text{ECI}}^{\text{MCI}} \cdot R_{\text{ECEF}}^{\text{ECI}} \begin{bmatrix} X_r \\ Y_r \\ Z_r \end{bmatrix} \tag{8}$$

where

$$R_{\text{ECEF}}^{\text{ECI}} = [\mathbf{P}][\mathbf{N}][\mathbf{\Theta}][\mathbf{\Pi}] \tag{9}$$

is the transformation matrix from ECEF to ECI. The rotation matrices $\mathbf{P}$, $\mathbf{N}$, $\mathbf{\Theta}$, and $\mathbf{\Pi}$ describe coordinate conversion due to precession, nutation, Earth rotation, and polar motion, respectively. Then,

$$R_{\text{ECI}}^{\text{MCI}} = R_3(0)R_1(0)R_3(0) + \begin{bmatrix} X_{\text{Moon}} \\ Y_{\text{Moon}} \\ Z_{\text{Moon}} \end{bmatrix}_{\text{ECI}} \tag{10}$$

is the transformation matrix from ECI to Moon-centered inertial (MCI). As shown in (1), the transformation is mainly determined by the Moon revolution. Also,

$$R_{\text{MCI}}^{\text{MCMF}} = R_3(-\Phi_m)R_1(-\theta_m)R_3(-\psi_m) \tag{11}$$

is the transformation matrix from MCI to MCMF. Φ_m, θ_m, and ψ_m are Euler angles, which can be obtained by calculating the JPL Development Ephemeris 430, in which the lunar libration is considered.

The coordinate of the lunar surface target in MCMF (X_t, Y_t, Z_t) can be obtained by

$$\begin{bmatrix} X_t \\ Y_t \\ Z_t \end{bmatrix} = \begin{bmatrix} a_m \cos \varphi_t \cos \lambda_t \\ a_m \cos \varphi_t \sin \lambda_t \\ a_m \sin \varphi_t \end{bmatrix} \tag{12}$$

where a_m is the lunar mean radius.

Consequently, based on the aforementioned analysis, the coordinates of the Earth-based radar in GCS and the lunar surface target in the selenographic coordinate system (SCS) are all transformed to MCMF, and the motion of the radar relative to the target can be calculated by

$$\vec{r} = \begin{bmatrix} X_m \\ Y_m \\ Z_m \end{bmatrix} - \begin{bmatrix} X_t \\ Y_t \\ Z_t \end{bmatrix}. \tag{13}$$

B. SRP Motion Trajectory

The SRP is the intersection of the lunar surface with a line between the radar and the lunar mass center. The SRP can characterize the relative position and attitude of the Earth and the Moon, and it is vital for investigating the relative motion model. As we know, the Moon always has the same face toward the Earth, where the Moon rotates around its axis with a period equal to its sidereal period of about 27 days, which is referred to as tidal locking [34]. Theoretically, the SRP would standstill because of the tidal locking effect. However, in fact, the SRP has been wagging around (0°E, 0°N) with the changing viewing perspective caused by libration.

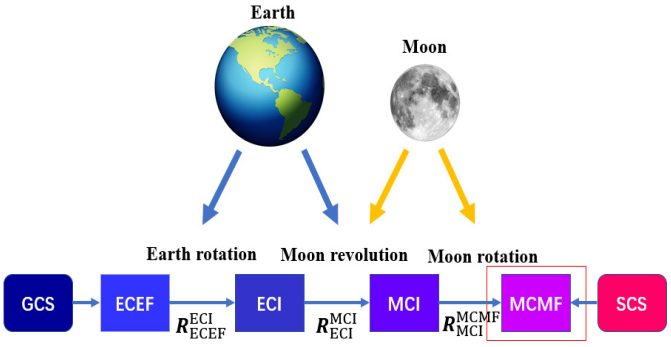


Fig. 3. Coordinate transformation rule.

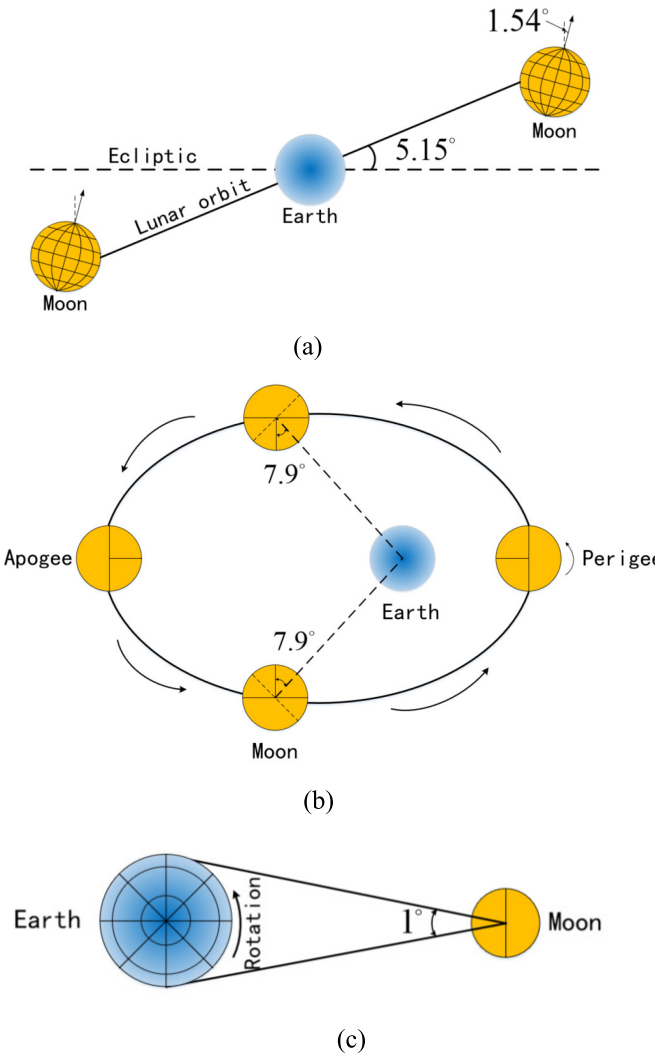


Fig. 4. Lunar libration. (a) Libration in latitude. (b) Libration in longitude. (c) Diurnal libration.

The latitude and longitude of the SRP, $\boldsymbol{\varphi}_{\mathbf{SRP}}$ and $\boldsymbol{\lambda}_{\mathbf{SRP}}$, are introduced to conveniently describe the SRP motion trajectory. The SRP wagging along the latitude dimension is mainly because the Moon's rotation axis is slightly inclined to the lunar orbit, which is also the reason why we have alternating

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views of the north and the south pole of the Moon in the course of the month. Moreover, as shown in Fig. 4(a), the SRP wagging amplitude along the latitude dimension can reach a maximum value $\delta_m = 6.7°$, which is the sum of the Moon's orbital inclination (5.15°) to the ecliptic and the negative equatorial tilt of 1.54°, and the angular velocity of the SRP along the latitude dimension is approximated by [42]

$$\dot{\boldsymbol{\varphi}}_{\mathbf{SRP}} = \delta_m \overline{\omega}_{\mathrm{rot}} \cos(\overline{\omega}_{\mathrm{rot}} t) \tag{14}$$

where $\overline{\omega}_{\mathrm{rot}}$ is the mean angular velocity of lunar rotation, and when $t = 0$, the latitude of SRP is 0° and $\dot{\boldsymbol{\varphi}}_{\mathbf{SRP}}$ reaches its maximum value of about 0.31×10^{-6} rad/s.

The SRP wagging along the longitude dimension is mainly due to the Moon's eccentric orbit. According to Kepler's third law, when the Moon is near the perigee, the Moon moves faster than its rotation speed, and the SRP moves eastward. When the Moon is near the apogee, the Moon moves slower than its rotation speed, and the SRP moves westward. As shown in Fig. 4(b), the amplitude can reach a maximum value of 7.9°, and the angular velocity of the SRP along the longitude dimension is approximated by [42]

$$\dot{\boldsymbol{\lambda}}_{\mathbf{SRP}} = \overline{\omega}_{\mathrm{rot}} \left[1 - \frac{(1 + \boldsymbol{e} \cos \boldsymbol{v})^2}{\left(1 - \boldsymbol{e}^2\right)^{\frac{3}{2}}} \right]. \tag{15}$$

As shown in (15), the first term in brackets accounts for the lunar rotation, whereas the second term accounts for the relative orbital motion. When $\boldsymbol{v} = 0°$ and $\boldsymbol{v} = 180°$, the Moon is at the perigee and perigee, respectively, and $\dot{\boldsymbol{\lambda}}_{\mathbf{SRP}}$ reaches the maximum values of -0.314×10^{-6} and 0.295×10^{-6} rad/s, respectively.

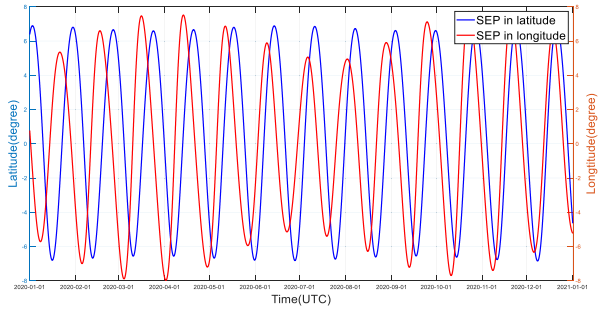
Besides, the Earth's rotation will also cause the movement of the SRP. As shown in Fig. 4(c), the SRP moving direction is perpendicular to the projection of the Earth's rotation axis on the Moon. With the rise and set of the Moon, the angular velocity of the SRP first increases and then decreases, and when crossing the sky, it can reach the maximum value in a day as [42]

$$\dot{\boldsymbol{\vartheta}}_{\mathbf{SRP}} = \frac{R_e \omega_e \cos \boldsymbol{\varphi}_{\boldsymbol{r}}}{|\vec{\boldsymbol{r}}|} \tag{16}$$

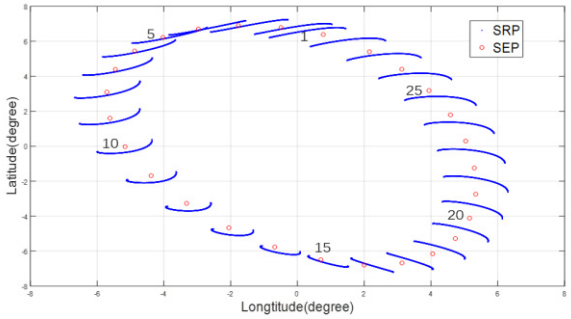
where R_e is the Earth's mean radius and ω_e is the mean angular velocity of Earth's rotation. When the radar is at the equator and the Moon is at the perigee, $\dot{\boldsymbol{\vartheta}}_{\mathbf{SRP}}$ has the maximum value and is approximately 1.2×10^{-6} rad/s.

According to the above analysis, we can conclude that the SRP motion has at least two periods. The first period is one day caused by the rotation of the Earth, and the other is about 27 days caused by the rotation and revolution of the Moon. Moreover, the SRP movement speed caused by the Earth's rotation is several times to dozens of times that caused by lunar rotation and revolution in the synthetic aperture time. In other words, the Earth's rotation is the main source to generate the synthetic aperture.

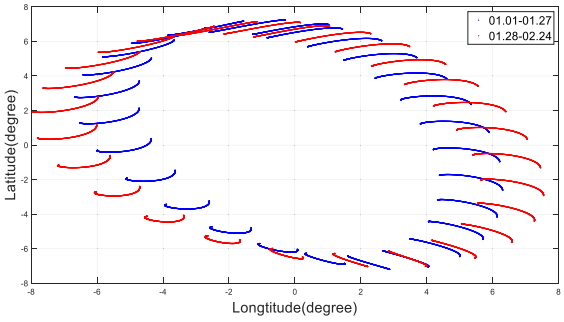
In addition, to simulate the real motion of the Earth and the Moon and describe the SRP motion trajectory accurately, based on the coordinates of the Earth-based radar in the



(a)



(b)



(c)

Fig. 5. SRP and SEP motion. (a) Variation of SEP in year 2020. (b) SRP and SEP motion trajectories in January 2020. (c) SRP trajectories of two adjacent sidereal months.

MCMF system (X_m, Y_m, Z_m), the latitude and longitude of the SRP can be expressed as

$$\boldsymbol{\varphi}_{\mathbf{SRP}} = \arctan\left(\frac{Z_m}{\sqrt{X_m^2 + Y_m^2}}\right) \tag{17}$$

$$\boldsymbol{\lambda}_{\mathbf{SRP}} = \arctan2(Y_m / X_m) \tag{18}$$

where arctan and arctan2 are all inverse tangent functions, arctan returns a value in the range from $-\pi/2$ to $\pi/2$, and arctan2 returns the four-quadrant value.

As shown in Fig. 5(a), the variation of sub-Earth point (SEP), which is the intersection of the lunar surface with a line between the Earth's mass center and the lunar mass center, in year 2020 is simulated. The SEP along the latitude dimension and the SEP along the longitude dimension change sinusoidally with an approximate period of about 27 days, which coincides with the previous analysis. However, due to

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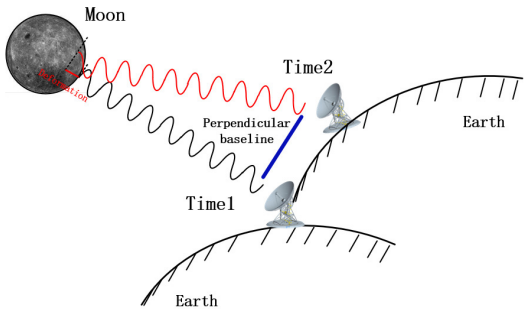


Fig. 6. Repeat-pass observation geometry of the Moon.

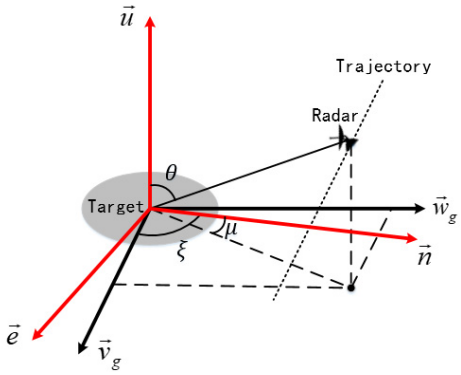


Fig. 7. Geometric model based on azimuth and incident angle.

the lunar precession, there is a little difference in the extrema of the latitudes and longitudes between months.

Meanwhile, assuming that the radar is located at (0°E, 0°N) on the Earth's surface, the SRP and SEP motion trajectories in January 2020 are also simulated, and the simulation result is shown in Fig. 5(b). Since the maximum longitude is out of sync with the maximum latitude, the SEP trajectory is an ellipse that is mainly decided by the argument of perigee. The SRP moves slowly with the SEP on a large time scale with a monthly cycle, and it also moves roughly to the west every day due to the Earth's rotation. Therefore, different from the spaceborne interferometric SAR system, Earth-based radar is more sensitive to the north–south deformation of the lunar surface [19]. The direction of SRP movement is perpendicular to the apparent rotation axis. It should be noted that the Doppler frequencies on both sides of the SRP are the same, so the SRP motion trajectory is also called the Doppler equator, which is also the reason for the north–south ambiguity in lunar imaging [42]. In order to avoid this situation, when observing the Moon, the radar beam should illuminate only one side of the Doppler equator at a time.

In addition, as shown in Fig. 5(c), the SRP trajectories of two adjacent sidereal months do not coincide, due to the axial precession, the apsidal precession, and the nodal precession. The complex motion of the Moon makes the libration circumstances change constantly, and no two lunations are the same, which means that every observation of the Moon is truly unique [26].

III. Coherence Analysis and Spatial–Temporal Baseline Calculation

A. Coherence Analysis

According to the abovementioned SRP wagging phenomenon, it can be seen that the orbits of the Earth-based radar during repeat-pass observation of the Moon change significantly, as shown in Fig. 6. Correspondingly, there is an obvious view angle difference of the same target during the repeat-pass observation, which will introduce phase noise in interferogram and reduce the coherence of the InSAR image pairs, namely, the geometric decoherence [38], and the geometric decoherence will make the InSAR processing fail in some cases.

Furthermore, temporal decorrelation, a main source of decoherence noise for Earth repeat-pass InSAR observations, can be ignored in the context of lunar due to the lack of atmosphere, weather, vegetation, and anthropogenic activity that are major sources of decorrelation on the Earth [21]. It should be noted that the Earth's atmospheric delay error is omitted here because it can be accurately corrected by existing methods [39]–[41]. Due to the narrow radar beam and the spotlight illumination mode, all targets in the imaging scene have almost the same phase history introduced by atmosphere and ephemeris errors. In other words, the difference of atmosphere and ephemeris errors between observations only affects the absolute deformation measurements, not the relative deformation. In addition, the topography is also a major source of defocus when processing to 5-m azimuth resolution [11]. The interferometric phase errors introduced by different autofocus parameters of the primary and secondary images can be avoided by the DEM-assisted time-domain backprojection algorithm (BPA).

1) Geometric Decorrelation Model Based on Azimuth and Incident Angle: To evaluate the effect of SRP wagging on InSAR coherence, a geometric decorrelation model based on azimuth and incident angle is proposed, as shown in Fig. 7. For the sake of simplicity, the topography variations on the Moon are neglected, and the Moon's surface is considered to be spherical. The normal direction to the lunar surface target can be written as

$$\vec{\boldsymbol{u}} = \left[\cos\varphi_t \cos\lambda_t \;\; \cos\varphi_t \sin\lambda_t \;\; \sin\varphi_t\right]^T \tag{19}$$

where the superscript T indicates matrix transposition. The unit vectors $\vec{\boldsymbol{e}}$ and $\vec{\boldsymbol{n}}$ in MCMF cartesian coordinates, which denote the local east direction and the local north direction, respectively, are given by

$$\vec{\boldsymbol{e}} = \left[-\sin\lambda_t \quad \cos\lambda_t \quad 0\right]^T \tag{20}$$

$$\vec{\boldsymbol{n}} = \left[-\cos\lambda_t \sin\varphi_t \quad -\sin\lambda_t \sin\varphi_t \quad \cos\varphi_t\right]^T. \tag{21}$$

The azimuth angles μ and ξ are the angles between the projection of radar line-of-sight (LOS) direction, and radar speed direction on the lunar surface and the north direction, respectively. $\vec{\boldsymbol{w}}_{\boldsymbol{g}}$ is the unit vector determined by $\vec{\boldsymbol{v}}_{\boldsymbol{g}} \times \vec{\boldsymbol{n}}$. Assuming that the resolution center point is at (0, 0, 0) in the $\vec{\boldsymbol{v}}_{\boldsymbol{g}} - \vec{\boldsymbol{w}}_{\boldsymbol{g}} - \vec{\boldsymbol{u}}$ coordinate framework, after SAR focusing, the SAR point spread function (PSF) on the ground can be

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expressed as

$$S = \iint W(x, y) \exp\left(-j\frac{4\pi f}{c} r(x, y)\right) \sigma(x, y) dx dy \quad (22)$$

where $W(x, y)$ is the SAR processor point target response, $\sigma(x, y)$ is the backscattering coefficient, f is the carrier frequency of the transmitted signal, (x, y) represents the scatter position in the ground resolution cell, and $r(x, y)$ is the slant range, which can be written as

$$r(x, y) = r_0 + x \cdot \sin\theta \cos(\xi - \mu) + y \cdot \sin\theta \sin(\xi - \mu) \quad (23)$$

where r_0 is the slant range from radar to the scattering center of the resolution cell. Then, the geometric coherence of repeat-pass InSAR signals can be calculated as [38]

$$\gamma_g = \frac{\left|\langle S_1 S_2^* \rangle\right|}{\sqrt{\langle S_1 S_1^* \rangle \langle S_2 S_2^* \rangle}} \approx \frac{\left|\iint |W(x, y)|^2 \exp\left(-j\left(\frac{4\pi f}{c}(\Delta r_0 + px + qy)\right)\right) dx dy\right|}{\left|\iint |W(x, y)|^2 dx dy\right|} \quad (24)$$

$$p = (\Delta\xi - \Delta\mu)\sin\theta \sin(\xi - \mu) - \Delta\theta \cos\theta \cos(\xi - \mu) \quad (25)$$

$$q = (\Delta\xi - \Delta\mu)\sin\theta \cos(\xi - \mu) + \Delta\theta \cos\theta \sin(\xi - \mu) \quad (26)$$

where S_1 and S_2 are SAR images obtained from two passes; superscript $*$ indicates complex conjugate; and Δr_0, $\Delta\theta$, $\Delta\xi$, and $\Delta\mu$ are the small differences of r_0, θ, ξ, and μ between repeat-pass observations, respectively.

According to [44], when the fractional bandwidth is no more than 0.1 and the integration angle is no more than 10°, the spectral supports are usually approximately rectangular and the corresponding PSF is usually approximately a 2-D Sinc function. In the situation of Earth-based radar imaging of the Moon, the integration angle is no more than 1° limited by the diurnal libration, and the fractional bandwidth is usually no more than 0.1 limited by radar transmitter power. Therefore, the PSF can be written as

$$W(x, y) = \mathrm{Sinc}\left(\frac{x}{\rho_x}\right)\mathrm{Sinc}\left(\frac{y}{\rho_y}\right). \quad (27)$$

Also, according to the Fourier transform of the Sinc function, the geometric coherence can be written as

$$\gamma_g = \left(1 - \frac{|p|}{\lambda/(2\rho_x)}\right) \cdot \left(1 - \frac{|q|}{\lambda/(2\rho_y)}\right) \quad (28)$$

where λ is the signal wavelength and ρ_x and ρ_y are azimuth resolution and ground range resolution, respectively. For small squint angle ($\xi - \mu \approx 90°$) and approximately parallel baseline ($\Delta\xi \approx 0°$), p and q can be simplified as [36]

$$p \approx \Delta\mu \sin\theta \quad (29)$$

$$q \approx \Delta\theta \cos\theta. \quad (30)$$

Thus, the geometric coherence can be rewritten as

$$\gamma_g = \left(1 - \frac{|\Delta\mu|}{\Delta\mu_c}\right) \cdot \left(1 - \frac{|\Delta\theta|}{\Delta\theta_c}\right) \quad (31)$$

where

$$\Delta\mu_c = \lambda/(2\rho_x \sin\theta) \quad (32)$$

$$\Delta\theta_c = \lambda/(2\rho_y \cos\theta) \quad (33)$$

are critical azimuth angle difference (AAD) and critical incident angle difference (IAD), respectively.

As shown in (31), the AAD and IAD of radar LOS during repeat-pass observation are the fundamental factors of decoherence, and the critical values are mainly determined by the radar wavelength, the size of resolution cell, and the radar incident angle as shown in (32) and (33). Assuming an incident angle of 30° and the resolution cell on the ground of about 20 m, the critical AAD and critical IAD at the S-band are about 0.29° and 0.17°, respectively. In practice, to ensure the interferometry viable, the geometric coherence should be greater than a suitable correlation threshold, which is set to 0.3 here. Then, the AAD and IAD should be no more than 0.2° and 0.12°, respectively, which are very strict due to the existence of SRP wagging phenomenon and the uncontrolled movement of the Earth and the Moon.

2) Quantitative Analysis of the Azimuth Angle and Incident Angle: To implement repeat-pass interferometry, it is demanding to investigate whether the view angle difference caused by the SRP wagging phenomenon, from day to day, satisfies the critical values. According to (13), (17), and (18), the slant range vector from the radar to the target can be rewritten as

$$\vec{\boldsymbol{r}} = \begin{bmatrix} r_m \cos\varphi_{\mathrm{SRP}} \cos\lambda_{\mathrm{SRP}} - a_m \cos\varphi_t \cos\lambda_t \\ r_m \cos\varphi_{\mathrm{SRP}} \sin\lambda_{\mathrm{SRP}} - a_m \cos\varphi_t \sin\lambda_t \\ r_m \sin\varphi_{\mathrm{SRP}} - a_m \sin\varphi_t \end{bmatrix} \quad (34)$$

where $r_m = (X_m^2 + Y_m^2 + Y_m^2)^{1/2}$ is the distance from the radar to the lunar center. Then, the azimuth angle of radar LOS can be written as

$$\mu = \arctan2\left(\frac{\vec{\boldsymbol{r}} \cdot \vec{\boldsymbol{e}}}{\vec{\boldsymbol{r}} \cdot \vec{\boldsymbol{n}}}\right) = \arctan2 \times \left(\frac{\cos\varphi_{\mathrm{SRP}} \sin(\lambda_{\mathrm{SRP}} - \lambda_t)}{\sin\varphi_{\mathrm{SRP}} \cos\varphi_t - \cos\varphi_{\mathrm{SRP}} \sin\varphi_t \cos(\lambda_{\mathrm{SRP}} - \lambda_t)}\right). \quad (35)$$

Subsequently, the incident angle can be calculated by

$$\theta = \arccos\left(\frac{\vec{\boldsymbol{r}}}{|\vec{\boldsymbol{r}}|} \cdot \vec{\boldsymbol{u}}\right) \approx \frac{r_m}{r_m - a_m}\alpha \quad (36)$$

where

$$\alpha = \arccos(\cos\varphi_{\mathrm{SRP}} \cos\varphi_t \cos(\lambda_{\mathrm{SRP}} - \lambda_t) + \sin\varphi_{\mathrm{SRP}} \sin\varphi_t) \quad (37)$$

is the selenocentric angle between SRP and lunar surface target, as shown in Fig. 8. The AAD and IAD between different passes can be approximated by the derivatives as

$$\Delta\mu = \frac{\partial\mu}{\partial\varphi_{\mathrm{SRP}}}\Delta\varphi_{\mathrm{SRP}} + \frac{\partial\mu}{\partial\lambda_{\mathrm{SRP}}}\Delta\lambda_{\mathrm{SRP}} \quad (38)$$

$$\Delta\theta = \frac{\partial\theta}{\partial r_m}\Delta r_m + \frac{\partial\theta}{\partial\varphi_{\mathrm{SRP}}}\Delta\varphi_{\mathrm{SRP}} + \frac{\partial\theta}{\partial\lambda_{\mathrm{SRP}}}\Delta\lambda_{\mathrm{SRP}} \quad (39)$$

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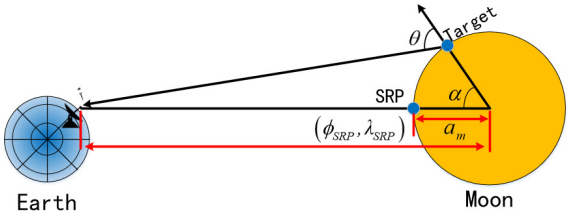


Fig. 8. Selenocentric angle between SRP and target.

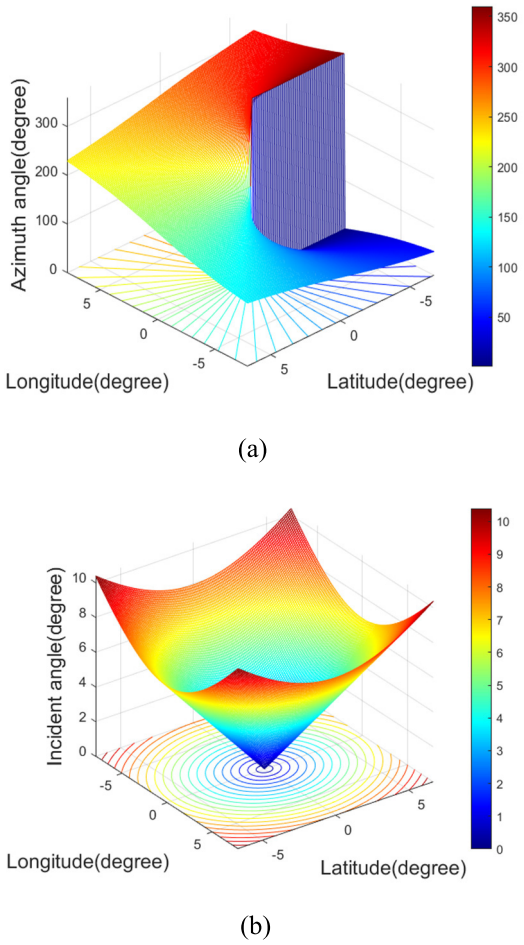


Fig. 9. (a) Azimuth angle and (b) incident angle with the changing of SRP position.

where Δr_m, $\Delta\varphi_{\mathrm{SRP}}$, and $\Delta\lambda_{\mathrm{SRP}}$ are the differences of slant range r_m, SRP latitude, and SRP longitude between the different passes, respectively. It can be seen that the AAD is only determined by $\Delta\varphi_{\mathrm{SRP}}$ and $\Delta\lambda_{\mathrm{SRP}}$, and the IAD is not only related to $\Delta\varphi_{\mathrm{SRP}}$ and $\Delta\lambda_{\mathrm{SRP}}$ but also related to Δr_m.

For specificity, assuming that the target is located at (0°E, 0°N) and the radar is 380 000 km away from the center of the Moon, the azimuth angle and incident angle with the changing of SRP position are simulated, and the results are shown in Fig. 9(a) and (b). As shown in Fig. 9(a), the azimuth angle is spiral-shaped with the changes of SRP longitude and SRP latitude. Moreover, according to (35), when $\lambda_{\mathrm{SRP}} = \lambda_t$, the azimuth angle equals 0° or 180°, which depends on the sign

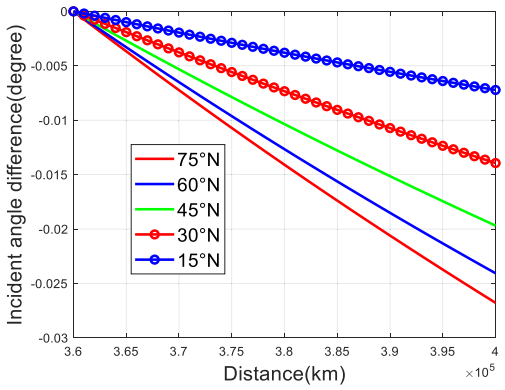


Fig. 10. IAD with the changing of distance.

of $\varphi_{\mathrm{SRP}} - \varphi_t$ and is consistent with Fig. 9(a). As shown in Fig. 9(b), the incident angle is funnel-shaped with the change of SRP longitude and latitude; according to (36) and (37), when the target position is the same as the SRP position, the incident angle has the minimum value 0°, and as SRP moves away from the target position, the incident angle gradually increases, which is consistent with Fig. 9(b).

Furthermore, the bottom of Fig. 9(a) and (b) shows the contour lines of the azimuth angle and the incident angle in the latitude–longitude plane, respectively. It should be noted that the contour lines can also indicate the gradient characteristics of the azimuth angle and incident angle: the gradient is always perpendicular to the contour lines, and the denser the contour lines are, the larger the gradient is. Thus, as shown in Fig. 9(a) and (b), because the azimuth angle contour lines and the incident angle contour lines are, respectively, radial and concentric, which are perpendicular to each other, the azimuth angle gradient and the incident angle gradient will be concentric shape and radial shape, respectively, which are also perpendicular to each other. What is more, along with the incidence angle increasing, the azimuth angle contours become sparse and the azimuth angle gradient becomes smaller. However, the incident angle gradient remains constant. Above all, for the same SRP variation, the larger the incident angle, the smaller the variation range of azimuth angle, and the stronger the coherence.

Besides, according to (39), the IAD is also related to the distance from the radar to the target. For specificity, assuming that the SRP is located at (0°E, 0°N), the IAF with the changing of the distance is simulated, and the results are shown in Fig. 10. It can be seen that the absolute value of the IAD approximately linearly increases with the distance, and the changing rate is determined by the location of the target. According to Fig. 8 and (36), the higher the latitude of the target, the larger the selenocentric angle, and the more significant the influence of distance change, which is consistent with Fig. 10. In addition, compared with the other two factors, the distance change has little effect on the IAD even for high-latitude targets, so it is ignored.

B. Spatial–Temporal Baseline Calculation

The previous part has quantitatively analyzed the relationships between the coherence and the incident angles and the

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TABLE II
CRITICAL BASELINE LENGTH

Parameter	Value
Incident angle (degree)	30
Earth-Moon distance (km)	380000

Critical Baseline Length (km)							
Band	Frequency (GHz)	Ground Range Resolution (m)					Wavelength (cm)
		3	5	10	20	50	
P	0.5	43800	26330	13160	6580	2630	60.0
L	1.5	14630	8780	4390	2190	880	20.0
S	3.0	7310	4390	2190	1100	440	10.0
C	6.0	3660	2190	1100	550	220	5.0
X	10.0	2190	1320	660	330	130	3.0
Ku	17.0	1290	770	390	190	80	1.8

TABLE III
INPUT PARAMETERS FOR CALCULATION

Parameters	Value	Unit
Radar frequency	3	GHz
Radar position on Earth	(0° N, 0° E)	/
Target position on lunar surface	Tycho (43.31° S, 11.36° W)	/
Earth's equatorial radius	6378.1370	km
Earth's polar radius	6356.7523	km
Moon's radius	1737.4	km
Date and Time	From 1 January 2020 to 31 December 2020	/
Synthetic aperture time	30	min
Critical AAD	0.29	degree

azimuth angels in detail, but the incident angle and azimuth angle are not convenient to evaluate the performance of the ground-based radar interferometry for the Moon. Thus, to address this problem, the spatial–temporal baseline is adopted here.

The Earth-based radar orbits the Moon at a distance of about 380 000 km and revisits the nearside on the Moon after a specific time. The period between two visits is called temporal baseline in InSAR. The distance between two acquisitions in a cross-track direction is known as the perpendicular baseline. The temporal baseline determines the sampling frequency of the deformation phase, and the spatial baseline determines the coherence and fringe density of interferograms. Therefore, the distribution of spatial–temporal baseline is the key to interferometric processing.

Nowadays, the temporal baseline of spaceborne repeat-pass InSAR is typically slightly more than ten days. For example, the temporal baselines of TerraSAR-X, Sentinel-1A, and ALOS-2 are 11 days, 12 days, and 14 days, respectively [43]. The revisit period of Earth-based radar for the near side of the Moon is only one day, which makes it possible to detect the rapid lunar surface changes and lunar mass wasting caused by moonquakes and asteroid impact and then answer the question of how active the lunar surface is now [46]. However, it should be noted that due to the influence of lunar libration and orbital precession, the interferometric processing cannot be carried out every day, which depends on the geometric coherence according to (31).

Moreover, as shown in Fig. 6, the perpendicular baseline of the repeat-pass observation can be calculated by

$$B_{\perp} = |\vec{\boldsymbol{r}}|\Delta\theta. \tag{40}$$

Due to the hundreds of thousands of kilometers distance between the Earth and the Moon, the critical baseline varies from tens of kilometers to tens of thousands of kilometers, which is related to radar frequency and ground range resolution, as shown in Table II.

IV. SIMULATION RESULTS AND DISCUSSION

The geometry model and theoretical framework are established in Sections II and III. Based on that, the spatial–temporal baseline for Earth-based repeat-pass InSAR of the Moon is simulated and analyzed in this section. Using the geometric decorrelation criteria, we examine the effects of radar frequency, position, and target location on the quantity of viable interferometric pairs.

A. Spatial–Temporal Baseline Distribution Characteristic

As mentioned previously, the JPL Development Ephemeris 430 data, in which the lunar libration and precession are considered, is used to simulate the relative motion of the Moon and the Earth. Table III summarizes the main physical parameters of the Earth and the Moon used for calculation. Considering the radar horizon and target visibility, the radar elevation should be above 0°. The synthetic aperture time can be approximately calculated by

$$T_{\text{syn}} = \frac{c}{2f\omega\rho_x} \tag{41}$$

where ω is the apparent rotation angular velocity, whose mean value is about 7.2×10^{-5} °/s. Assuming that the azimuth resolution is about 20 m, the synthetic aperture time is about 30 min at S-band.

To evaluate the effective interferometric baselines from day to day, the change regularity of the AAD and IAD is investigated within a year. The minimum values of the average AAD and IAD in a synthetic aperture time every day between the primary acquisition date and the secondary acquisition date are calculated, as shown in Fig. 11(a) and (b), respectively. It can be seen that the AAD and IAD have the following similar distribution characteristics.

1) The AAD and IAD have a period of about 27 days, which coincides with the lunar libration period. Also, there are about 13.5 cycles in a year.
2) At a given primary acquisition day number, the AAD and IAD change significantly with the secondary acquisition day number in a form of absolute Sine function, as shown in the left of Fig. 11(a) and (b), and the peak values vary with different primary acquisition day number as the same form, as shown in the top of Fig. 11(a) and (b).
3) The AAD and IAD can reach up to 18° and 15°, respectively, which are much larger than the critical

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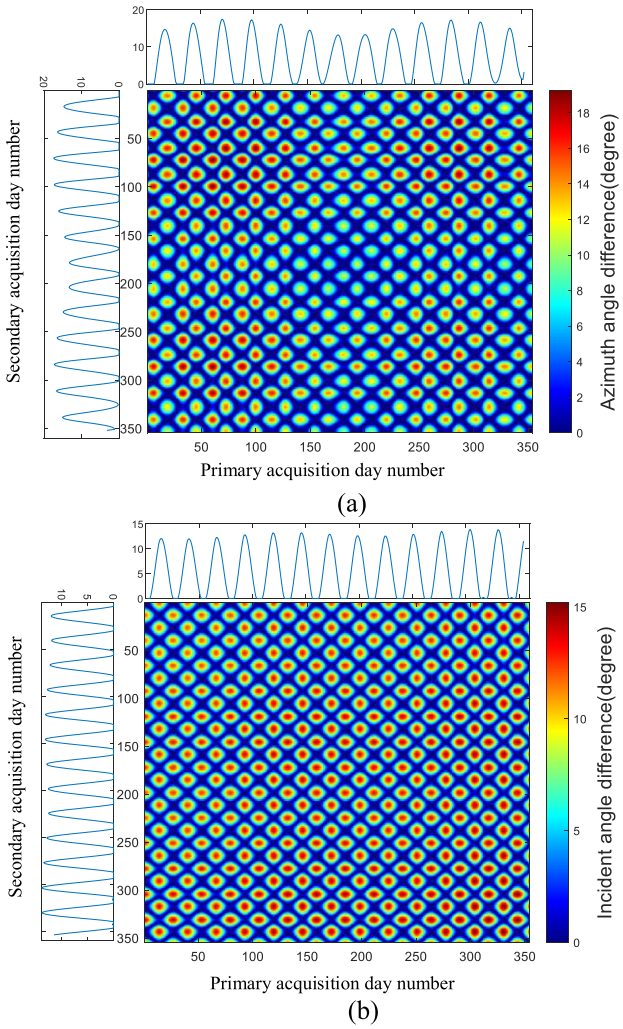


Fig. 11. AAD and IAD within a year. (a) AAD: the left and top sides are slices along the vertical axis and transverse axis, respectively. (b) IAD: the left and top sides are slices along the vertical axis and transverse axis, respectively.

value aforementioned. Also, the minimum values usually appear on diagonal lines of the 2-D ordination diagram.

Furthermore, the AAD is not exactly synchronized with the IAD. This is mainly because the periods of the AAD and the IAD are slightly different, which can be seen from the end phase of absolute Sine function wave in the top of Fig. 11(a) and (b). In other words, the AAD and IAD do not necessarily get the minimum value at the same time, which is unfavorable for baseline selection and interferogram coherence according to (31).

On the premise that the AAD is less than the critical value, the spatial–temporal baseline is analyzed, and the statistical distribution result is shown in Fig. 12. Moreover, in Fig. 12, the length of the spatial baseline is expressed in dB of kilometers, and the color represents the number of corresponding temporal–spatial baseline in the bin whose size is 1 day $\times$ 1.25 km.

As shown in Fig. 12, the temporal baseline has obvious periodicity, and the cycle is about 27 days. When the temporal baseline is an integral multiple of 27, such as 27 days and 54 days, the AAD and IAD usually have the minimum value according to Fig. 11(a) and (b), and the premise that the AAD is less than the critical value is easier to meet. Therefore, the spatial baseline reaches the local minimum value and the number of baselines reaches the local maximum value at the same time, which would benefit the repeat-pass Moon observation.

Furthermore, according to the statistical histogram of spatial baseline, the spatial baseline varies greatly from 1 to 100 000 km, and the shorter the baseline, the smaller the number. According to Table II, the critical spatial baselines are about 6580, 1100, and 330 km when the ground range resolution is 20 m in P-band, S-band, and X-band, as shown in the red, yellow, and black dotted lines in Fig. 12, respectively. There are about 7000, 1100, and 400 pairs of baselines are within the critical baseline in P-band, S-band, and X-band, respectively, as shown in Fig. 12, and as the critical baseline increases, the number of effective baselines will surge.

Correspondingly, according to the statistical histogram of the temporal baseline, there is a high possibility that the temporal baseline is only one day, which would significantly benefit the repeat-pass InSAR observation.

B. Effective Baseline Formation Frequency

To quantitatively evaluate the ability of Earth-based repeat-pass InSAR of the Moon from the aspect of baseline formation, the evaluation index of effective baseline formation frequency (EBFF) is adopted, and it can be expressed as

$$\text{EBFF} = \frac{M}{N^2} \tag{42}$$

where M is the effective baseline number over a length of time and N is the number of cycles. According to (42), the larger the EBFF, the more effective baselines are formed between a primary acquisition cycle and a secondary acquisition cycle.

Using the geometric decorrelation criteria, we examine the effects of radar frequency, position, and target location on the quantity of viable interferometric pairs.

1) Influence of Radar Frequency on EBFF: Radar power, radar bandwidth, and radar frequency are three key parameters, which have bind effects mutually and together determine the radar performance. Generally, the radar bandwidth increases with the increase of frequency in spaceborne SAR [43], and the ratio of radar bandwidth to frequency is the deciding factor of the interferometric critical baseline [21]. However, since the Earth-based radar is very far from the Moon, the radar power limits the increase of bandwidth. Usually, the radar power is limited and constant, and larger bandwidth means finer range resolution and lower signal-to-noise ratio (SNR) [35]. Therefore, for the sake of simplicity, assuming that the radar bandwidth and resolution both remain constant, the radar frequency is the deciding factor of critical value according to (32) and (33).

Assuming that the azimuth resolution and ground range resolution are 20 m, the EBFF in P-, L-, S-, C-, X-, and Ku-bands is simulated, and the results are shown in Fig. 13. It can be seen that the EBFF decays with the radar frequency exponentially. According to Table II, the lower the

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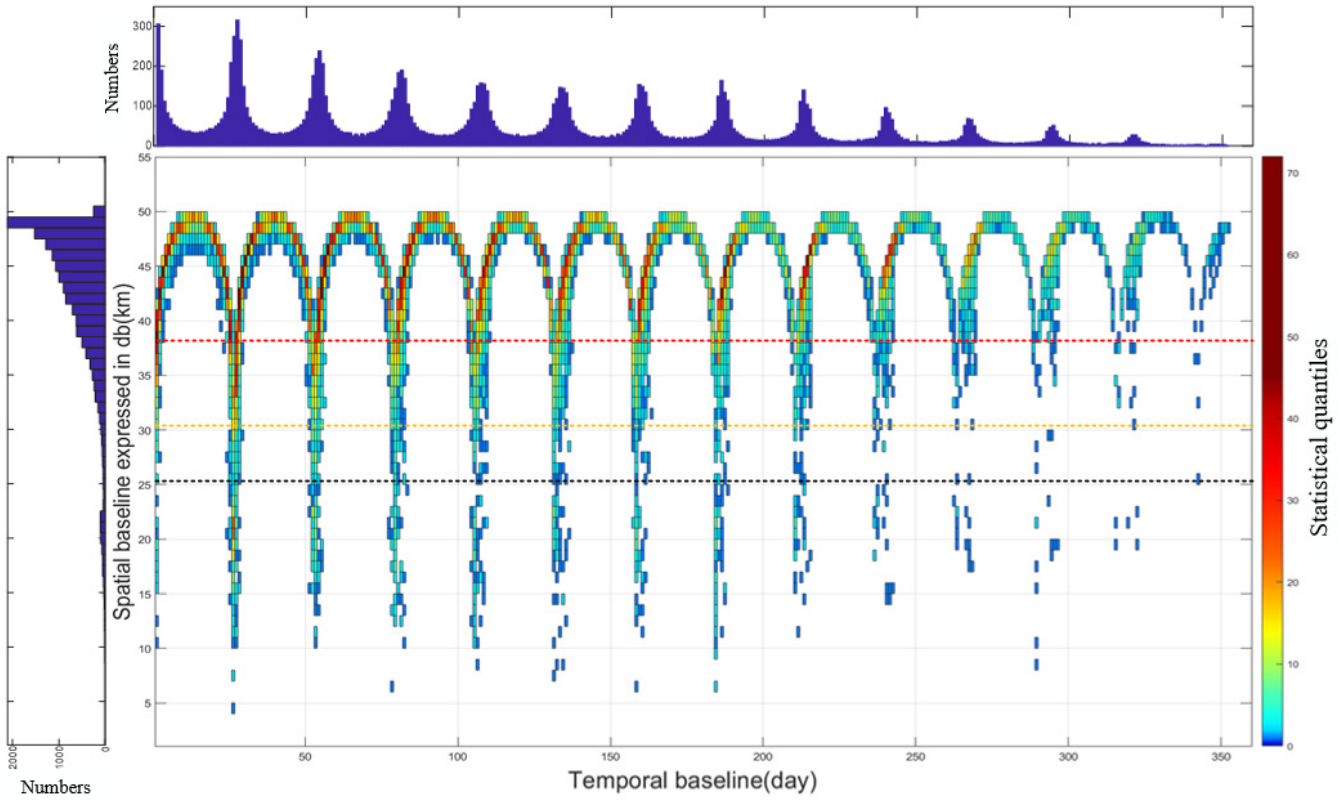


Fig. 12. Statistical distribution of spatial–temporal baseline: the left side is the statistical histogram of spatial baseline and the top side is the statistical histogram of temporal baseline.

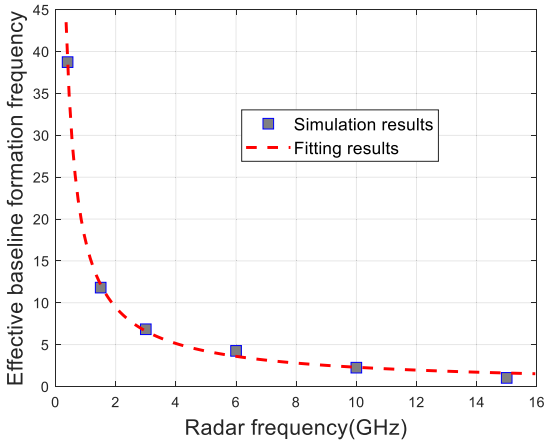


Fig. 13. Effects of radar frequency on the EBFF.

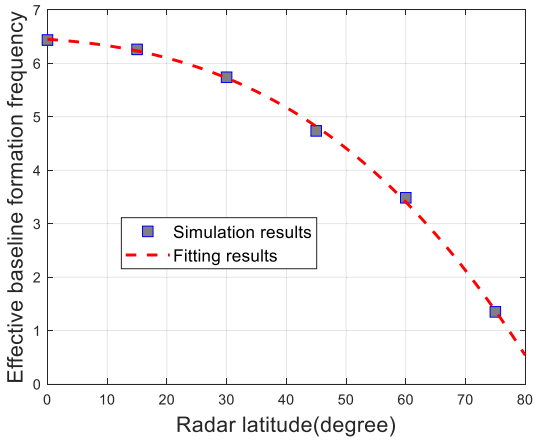


Fig. 14. Effects of radar latitude on the EBFF, and the radar longitude is 0°E.

radar frequency, the longer the critical baseline, so it is more conducive to form the viable interferometric baseline. According to Fig. 13, the EBFF is significantly affected by the radar frequency in the P- and S-bands and little in the C- and Ku-bands. When the radar frequency is above the Ku-band, the EBFF tends to converge. From the aspect of the number of effective baselines, the low-frequency band is preferred. However, the volume scattering decorrelation caused by low-frequency penetration will also degrade the InSAR performance [37], [38]. Thus, equally considering this phenomenon, S-band may be a better choice.

2) Influence of Radar Position on EBFF: The influence of radar latitude on EBFF is shown in Fig. 14. The EBFF roll-off is with the increase of radar latitude, and the higher the latitude, the sharper the drop-off. This is mainly because the angular velocity of the SRP at low latitude is larger according to (16), which means that the range of SRP motion is larger during daily observation and is more conducive to baseline formation. It should be noted that the radar latitude also leads to the change of SRP position, but the maximum change is no more than 1°, so its corresponding effect is limited.

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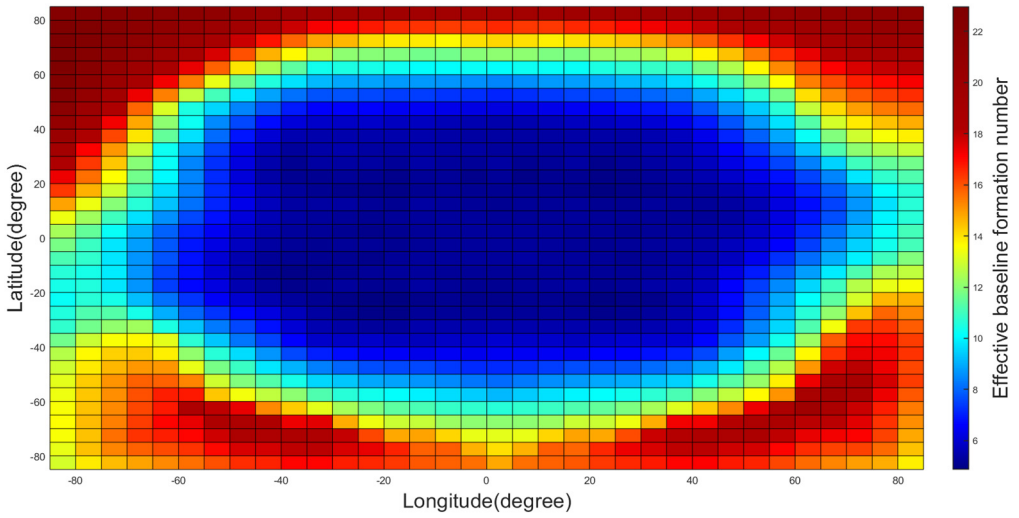


Fig. 15. Effects of lunar target position on the EBFF.

As shown in Fig. 14, when the radar latitude is less than 30°, at least five pairs of effective baselines can be formed during any two cycles. Therefore, when selecting the radar position, the low-latitude area should be mainly considered.

3) Influence of Lunar Target Position on EBFF: The influence of lunar target on EBFF is simulated, and the results are shown in Fig. 15. It can be seen that the EBFF grows with the lunar target latitude and longitude. Since the variation range of SRP is limited to low latitude (−7°∼7°) and low longitude (−8°∼8°), the target at high latitude (longitude) is usually far from the SRP and has a larger incident angle and smaller azimuth angle gradient based on the aforementioned analysis in Section III-A2. It means that the higher the target latitude, the larger the critical IAD according to (33), and the smaller the variation of azimuth angle for the same SRP variation. Therefore, according to (31), the higher the target latitude (longitude), the stronger the coherence, and, hence, the greater the EBFF.

It should be noted that the EBFF may decrease slightly due to the reduction of visible time at some very high latitude and longitude in Fig. 15. In particular, when the target is near the lunar polar region, more than 15 pairs of viable baselines can be formed. Due to such a high revisit frequency, it may be excepted to provide clues for the temporal variability of frost and ice water in seasonally shadowed regions [45] and monitor the health status of the future lunar base through the Earth-based repeat-pass InSAR technology with long term.

V. Conclusion

This article analyzed the potentialities of the Earth-based repeat-pass InSAR of the Moon from the aspect of spatial–temporal baseline formation.

The baseline formation mechanism was first investigated. Theoretical analyses showed that the interferometric baselines mainly come from the lunar libration; however, the lunar libration may also result in significantly changing baselines and further lead to serious geometric decorrelation. To address this problem and ensure Earth-based repeat-pass InSAR of the Moon, the precise relative motion model between the radar and the lunar target, considering lunar libration, was established, and the SRP wagging phenomenon was analyzed.

Then, to evaluate the effects of SRP wagging on InSAR coherence, a geometric decorrelation model based on azimuth angle and incident angle was proposed. Based on that, the spatial–temporal baseline for Earth-based repeat-pass InSAR of the Moon is simulated and analyzed. The presented results show that the temporal baseline has obvious periodicity with a cycle of about 27 days. When the temporal baseline is an integral multiple of 27, the spatial baseline usually has the minimum value and the geometric decorrelation effect is relatively slight.

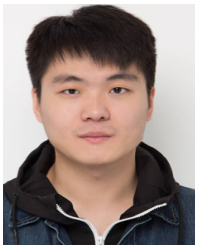
Moreover, the evaluation index of EBFF was proposed to quantitatively evaluate the ability of Earth-based repeat-pass InSAR of the Moon from the aspect of baseline formation. Also, the effects of radar frequency, radar position, and lunar target position on the EBFF were discussed, which has guiding significance for the selection of radar system parameters, radar position, and lunar target area. It was found that the low-frequency and the low-latitude radar are conducive to the baseline formation. Also, in the high latitudes of the Moon, especially the polar regions, the Earth-based repeat-pass InSAR technology is expected to play a potentially great role with a high acquisition frequency.

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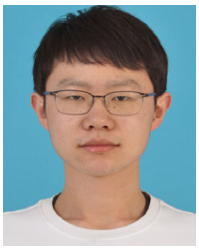
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